Ratsional, irratsional va haqiqiy sonlar

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Nechtasi irratsional: $\sqrt{36}$; $\pi - 3$; 0,(12); $\sqrt{12}$; $\frac{22}{7}$; $\sqrt[3]{-1}$?**

A) 1 **B)** 2 **C)** 3 **D)** 4

2. **Eng kichik sonni toping: −0,(72); −0,7(2); −0,72; −0,72(7).**

A) −0,72 **B)** −0,(72) **C)** −0,72(7) **D)** −0,7(2)

3. **Qaysi tengsizlik to'g'ri?**

A) $\pi < 3{,}14$ **B)** $3{,}14 < \pi < \frac{22}{7}$

C) $\frac{22}{7} < \pi$ **D)** $\pi = \frac{22}{7}$

4. **$\frac{3}{7}$ ning o'nli yozuvida davrda nechta raqam bor?**

A) 2 **B)** 3 **C)** 7 **D)** 6

5. **$\frac{7}{12}$ ni o'nli kasr ko'rinishida yozing.**

A) 0,(583) **B)** 0,583 **C)** 0,5(83) **D)** 0,58(3)

6. **$-1\frac{5}{6}$ ni o'nli kasr ko'rinishida yozing.**

A) −1,83 **B)** −1,(83) **C)** −1,8(3) **D)** −0,8(3)

7. **$\frac{1}{90} = 0{,}0(1)$ dan foydalanib, 0,8(3) ni oddiy kasr ko'rinishida yozing.**

A) $\frac{83}{90}$ **B)** $\frac{8}{3}$ **C)** $\frac{5}{6}$ **D)** $\frac{83}{100}$

8. **0,3(5) ni oddiy kasr ko'rinishida yozing.**

A) $\frac{35}{99}$ **B)** $\frac{16}{45}$ **C)** $\frac{7}{20}$ **D)** $\frac{35}{100}$

9. **$\frac{1}{5}$ va $\frac{1}{4}$ orasidagi qaysi son irratsional?**

A) 0,2(1) **B)** 0,2101001000…

C) 0,22 **D)** 0,2(3)

10. **Qaysi tasdiq yolg'on?**

A) Har qanday natural son — haqiqiy son

B) Irratsional sonning kvadrati har doim irratsional son

C) $\pi + 4$ — irratsional son

D) Har qanday ratsional son chekli yoki davriy o'nli kasr bo'ladi

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11. **«Ratsional son bilan irratsional sonning ko'paytmasi har doim irratsional» tasdig'iga qaysi juft kontrmisol bo'ladi?**

A) 0 va $\sqrt{2}$ **B)** 2 va $\sqrt{2}$

C) -1 va π **D)** $\sqrt{2}$ va $\sqrt{2}$

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12. **Rasmda A, B, C nuqtalar $\frac{22}{7}$, π va 3,1(5) sonlarini ifodalaydi (tartibsiz). B nuqta qaysi son?**

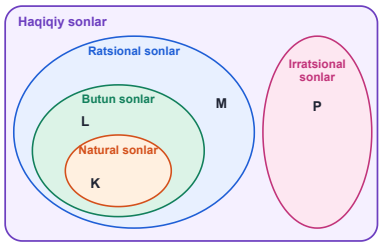


A) $\frac{22}{7}$ **B)** 3,14

C) π **D)** 3,1(5)

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13. **Rasmda sonlar to'plamlari, sohalarda harflar. Qaysi son M sohaga tushadi?**



A) $-1,(6)$ **B)** $\sqrt{7}$

C) $-\sqrt{9}$ **D)** $\sqrt{16}$

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14. **Rasmda 2 ni 13 ga bo'lishdagi qoldiqlar va bo'linma raqamlari. $\frac{2}{13}$ qanday yoziladi?**



A) 0,153846 **B)** 0,(153846)

C) 0,(15384) **D)** 0,1(53846)

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15. **Qaysi davriy kasr $-\frac{1}{4}$ va $-\frac{1}{5}$ orasida?**

A) $-0,1(6)$ **B)** $-0,2(5)$ **C)** $-0,2(3)$ **D)** $-0,(3)$

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16. **Manfiy ratsional son $\frac{3}{4}$ ga ham, $\frac{3}{8}$ ga ham bo'linganda butun son chiqadi. Uning eng katta qiymati qaysi?**

A) -1 **B)** $-\frac{3}{32}$ **C)** $-\frac{3}{4}$ **D)** $-\frac{3}{8}$

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17. $\left(-\frac{4}{3}\right)^3$ **ni o'nli kasr ko'rinishida yozing.**

A) $-2,3(70)$ **B)** $-2,(370)$ **C)** $-2,370$ **D)** $-2,(37)$

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18. **Qaysi kasr davriy o'nli kasrga aylanadi?**

A) $\frac{14}{21}$ **B)** $\frac{9}{24}$ **C)** $\frac{6}{15}$ **D)** $\frac{7}{35}$

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19. $\frac{1}{7}$ **= 0,(142857).** $\frac{2}{7}$ **ning o'nli yozuvi qaysi?**

A) 0,(28) **B)** 0,(285714)

C) 0,2(85714) **D)** 0,(284714)

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20. $\sqrt{30}$ **haqida qaysi gap to'g'ri?**

A) 6 bilan 7 orasida, irratsional

B) 15 bilan 16 orasida, irratsional

C) 5 bilan 6 orasida, ratsional

D) 5 bilan 6 orasida, irratsional

Javoblar

Ratsional, irratsional va haqiqiy sonlar · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	2
2	**C**	−0,72(7)
3	**B**	$3{,}14 < \pi < \frac{22}{7}$
4	**D**	6
5	**D**	0,58(3)
6	**C**	−1,8(3)
7	**C**	$\frac{5}{6}$
8	**B**	$\frac{16}{45}$
9	**B**	0,2101001000…
10	**B**	Irratsional sonning kvadrati har doim irratsional son

№	Javob	Qiymati
11	**A**	0 va $\sqrt{2}$
12	**A**	$\frac{22}{7}$
13	**A**	−1,(6)
14	**B**	0,(153846)
15	**C**	−0,2(3)
16	**C**	$-\frac{3}{4}$
17	**B**	−2,(370)
18	**A**	$\frac{14}{21}$
19	**B**	0,(285714)
20	**D**	5 bilan 6 orasida, irratsional

Qisqa yechimlar

1. $\sqrt{36} = 6$, 0,(12) — davriy, $\frac{22}{7}$ — kasr, $\sqrt[3]{-1} = -1$: ratsional.
$\pi - 3 = 0{,}14159\ldots$ va $\sqrt{12}$ (12 to'la kvadrat emas) — irratsional. Javob: **2**.

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2. Modullarini bir xil xonagacha yozamiz: 0,(72) = 0,72727…, 0,7(2) = 0,72222…, 0,72 = 0,72000, 0,72(7) = 0,72777…
Manfiy sonlarda moduli katta son kichik. Eng katta modul — 0,72777…: eng kichik son **−0,72(7)**.

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3. $\pi = 3{,}14159\ldots$, $\frac{22}{7} = 3{,}142857\ldots$
3-xonada $1 < 2$, demak $\pi < \frac{22}{7}$; 3,14 esa π dan kichik. Javob: $3{,}14 < \pi < \frac{22}{7}$.

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4. 7 ga bo'lganda qoldiq 1 dan 6 gacha — ko'pi bilan 6 xil.
3 : 7 da qoldiqlar 3, 2, 6, 4, 5, 1 — hammasi chiqadi: $\frac{3}{7}$ = 0,(428571). Javob: **6**.

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5. 7 : 12 = 0,58333… — qoldiqlar 7, 10, 4, 4, … 4 takrorlandi.
Takrorlanadigan faqat 3: **0,58(3)**.

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6. Butun qism va ishora alohida: 5 : 6 = 0,8333… = 0,8(3).
$-1\frac{5}{6} =$ **−1,8(3)**.

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7. 0,8(3) − 0,8 = 0,0(3) = $3 \cdot \frac{1}{90} = \frac{3}{90}$.
0,8(3) = $\frac{8}{10} + \frac{3}{90} = \frac{72}{90} + \frac{3}{90} = \frac{75}{90} = \frac{5}{6}$.
Tekshirish: 5 : 6 = 0,833…

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8. 0,3(5) − 0,3 = 0,0(5) = $5 \cdot \frac{1}{90} = \frac{5}{90}$.
0,3(5) = $\frac{3}{10} + \frac{5}{90} = \frac{27}{90} + \frac{5}{90} = \frac{32}{90} = \frac{16}{45}$.
($\frac{7}{20} = \frac{35}{100}$ = 0,35 — chekli kasr, boshqa son.)

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9. 0,2(1) va 0,2(3) — davriy, 0,22 — chekli: ratsional.
0,2101001000… da nollar har safar bittaga ko'payadi — takrorlanadigan guruh yo'q: **irratsional**.

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10. Kontrmisol: $\sqrt{3}$ — irratsional, lekin $(\sqrt{3})^2 = \sqrt{3} \cdot \sqrt{3} = 3$ — ratsional. Demak bu tasdiq **yolg'on**.
Qolgan uchtasi rost.

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11. Kontrmisol — biri ratsional, biri irratsional, ko'paytmasi esa ratsional bo'lishi kerak.
$0 \cdot \sqrt{2} = 0$ — ratsional: **0 va** $\sqrt{2}$.
$2\sqrt{2} = 2{,}828\ldots$ va $-\pi$ — irratsional; $\sqrt{2}$ va $\sqrt{2}$ — ikkalasi irratsional, shartga mos emas.

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12. Bir xil xonagacha: $\pi = 3{,}1415\ldots$, $\dfrac{22}{7} = 3{,}1428\ldots$, 3,1(5) = 3,1555…
O'sish tartibi: π, $\dfrac{22}{7}$, 3,1(5) — A, B, C. B — $\dfrac{22}{7}$.

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13. M — ratsional, lekin butun emas.
$\sqrt{16} = 4$ — natural (K), $-\sqrt{9} = -3$ — butun (L), $\sqrt{7}$ — irratsional (P).
−1,(6) — davriy kasr, butun emas: **M**.

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14. Qoldiqlar 2, 7, 5, 11, 6, 8 va yana 2 — birinchi qoldiq qaytdi.
Demak davr birinchi raqamdan boshlanadi va 6 ta raqam: **0,(153846)**.

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15. $-\dfrac{1}{4} = -0{,}25$, $-\dfrac{1}{5} = -0{,}2$.
−0,2(3) = −0,2333… — orasida. −0,(3) = −0,333… va −0,2(5) = −0,2555… — −0,25 dan kichik, −0,1(6) = −0,1666… — −0,2 dan katta. Javob: **−0,2(3)**.

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16. Son $\dfrac{3}{4}$ ning butun songa ko'paytmasi bo'lishi kerak: $-\dfrac{3}{4}, -\dfrac{6}{4}, \ldots$ Eng kattasi $-\dfrac{3}{4}$.
Tekshiramiz: $-\dfrac{3}{4} : \dfrac{3}{4} = -1$, $-\dfrac{3}{4} : \dfrac{3}{8} = -2$. Javob: $-\dfrac{3}{4}$.
$-\dfrac{3}{8} : \dfrac{3}{4} = -\dfrac{1}{2}$ — butun emas.

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17. $\left(-\dfrac{4}{3}\right)^3 = -\dfrac{64}{27}$ (toq daraja — ishora manfiy).
64 : 27 = 2, qoldiq 10; keyin qoldiqlar 10, 19, 1, 10 — 10 qaytdi: 2,370370… Javob: **−2,(370)**.

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18. Avval qisqartiramiz: $\dfrac{7}{35} = \dfrac{1}{5}$ = 0,2; $\dfrac{9}{24} = \dfrac{3}{8}$ = 0,375; $\dfrac{6}{15} = \dfrac{2}{5}$ = 0,4 — chekli.
$\dfrac{14}{21} = \dfrac{2}{3}$ = 0,(6) — davriy. Javob: $\dfrac{14}{21}$.

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19. $\dfrac{2}{7} = 2 \cdot \dfrac{1}{7}$: 2 : 7 da qoldiqlar 2, 6, 4, 5, 1, 3 — davr 6 raqam: 0,285714…
Tekshirish: 0,142857 · 2 = 0,285714. Javob: **0,(285714)**.

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20. $25 < 30 < 36$, demak $5 < \sqrt{30} < 6$.
30 to'la kvadrat emas — $\sqrt{30} = 5{,}477\ldots$ irratsional. Javob: **5 bilan 6 orasida, irratsional**.
